Demand Model Specification

EMS Optimization Project - Phase 2: Demand Modeling

Last Updated: March 12, 2026

Executive Summary

This document specifies the stochastic demand models fitted for crash arrivals in the EMS discrete-event simulation. Based on 416,434 Manhattan crash records (2012-2026), we developed:

1. **Baseline Homogeneous Poisson** - Constant rate model (rejected by chi-square test)
2. **Non-Homogeneous Poisson Process (NHPP)** - Recommended model with hourly and day-of-week variation
3. **Spatial Demand Model** - Precinct-level arrival rates

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1. Data Summary

Metric	Value
Total crashes	416,434
Date range	July 1, 2012 - February 20, 2026
Duration	4,983 days (119,592 hours)
Average daily rate	83.6 crashes/day
Average hourly rate	3.48 crashes/hour

Geographic Distribution

- **CBD crashes**: 231,786 (55.7%)
- **Non-CBD crashes**: 184,648 (44.3%)

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2. Homogeneous Poisson Model

Model Specification

$$N(t) \sim \text{Poisson}(\lambda t)$$

Where:
- λ = 3.482 crashes/hour (constant)

Goodness-of-Fit Test

Test	Statistic	p-value	Result
Chi-square	723,713.09	< 0.0001	**REJECT**
Dispersion (Var/ Mean)	2.43	-	Overdispersed

Conclusion: The homogeneous Poisson assumption is **rejected**. The data shows significant time-varying patterns that require a non-homogeneous model.

Limitations

- Ignores clear hourly and daily patterns
- Underestimates demand during peak hours
- Overestimates demand during off-peak hours
- Not suitable for realistic simulation

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3. Non-Homogeneous Poisson Process (NHPP)

Model Specification

The recommended model uses a time-varying arrival rate:

$$\lambda_{base} \times f_{hour}(h) \times f_{dow}(d)$$

Where:
- **λ_base** = 3.482 crashes/hour (overall mean rate)
- **f_hour(h)** = hourly factor for hour h ∈ {0, 1, …, 23}
- **f_dow(d)** = day-of-week factor for day d ∈ {0=Mon, 1=Tue, …, 6=Sun}

3.1 Hourly Factors

Hour	λ (crashes/hr)	Factor	Description
0	3.46	0.88	Midnight
1	2.42	0.61	
2	2.08	0.53	
3	1.93	0.49	
4	1.94	0.49	
5	1.78	**0.45**	**Minimum**
6	2.17	0.55	
7	2.61	0.66	Morning ramp-up
8	4.24	1.08	Morning commute
9	4.81	1.22	
10	4.83	1.23	
11	5.03	1.28	
12	5.15	1.31	Noon
13	5.27	1.34	
14	5.76	1.47	Afternoon
15	5.20	1.32	
16	5.97	**1.52**	**Peak**
17	5.70	1.45	Evening commute
18	5.28	1.34	
19	4.60	1.17	
20	4.04	1.03	
21	3.60	0.91	
22	3.44	0.87	
23	3.10	0.79	

Peak Period: 2 PM - 6 PM (factors > 1.3)
Low Period: 3 AM - 6 AM (factors < 0.5)

3.2 Day-of-Week Factors

Day	λ (crashes/day)	Factor
Monday	78.2	0.94
Tuesday	86.4	1.03
Wednesday	87.1	1.04
Thursday	90.3	1.08
Friday	**95.8**	**1.15**
Saturday	79.4	0.95
Sunday	**67.9**	**0.81**

Key Insight: Friday has 41% more crashes than Sunday.

3.3 Combined Rate Calculation

Example: Friday at 5 PM

```
λ(Friday, 17:00) = 3.482 × 1.45 × 1.15 = 5.81 crashes/hour
```

Example: Sunday at 5 AM

```
λ(Sunday, 05:00) = 3.482 × 0.45 × 0.81 = 1.27 crashes/hour
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4. Spatial Demand Model

Precinct-Level Rates

Precinct	Crashes	λ/hour	λ/day	% Total	Category
19 (Midtown East)	41,097	0.344	8.25	9.9%	**High**
18 (Midtown North)	35,577	0.297	7.14	8.5%	**High**
14 (Midtown South)	31,551	0.264	6.33	7.6%	**High**
1 (Financial District)	27,999	0.234	5.62	6.7%	**High**
17 (Midtown East)	27,628	0.231	5.54	6.6%	**High**
13 (Lower East Side)	25,101	0.210	5.04	6.0%	**High**
10 (Chelsea)	23,796	0.199	4.78	5.7%	**High**
…	…	…	…	…	…
22 (Central Park)	4,209	0.035	0.84	1.0%	Low
50 (Marble Hill)	765	0.006	0.15	0.2%	Low
52 (Inwood)	284	0.002	0.06	0.1%	Low
114 (Randall's Island)	80	0.001	0.02	0.0%	Low

High-Demand vs Low-Demand Zones

Category	Precincts	Combined λ/day
High (≥75th percentile)	19, 18, 14, 1, 17, 13, 10	46.7 (55.9%)
Low (≤25th percentile)	28, 30, 26, 22, 50, 52, 114	5.0 (6.0%)

5. CBD vs Non-CBD Patterns

Overall Rates

Area	λ (crashes/hour)	λ (crashes/day)	% of Total
CBD	1.94	46.5	55.7%
Non-CBD	1.54	37.1	44.3%

Temporal Pattern Differences

Characteristic	CBD	Non-CBD
Peak hour	14:00 (2 PM)	16:00 (4 PM)
Peak factor	1.60	1.71
Minimum hour	4 AM	3 AM
Peak day	Friday	Friday
Weekend dropoff	Sharper	Moderate

Insight: CBD peaks earlier (business hours) while non-CBD peaks during evening commute.

6. Implementation for Simulation

Arrival Generation Algorithm

```python
import numpy as np
import pandas as pd

# Load rate tables
hourly_factors = pd.read_csv('demand_lambda_hourly.csv').set_index('hour')['factor']
dow_factors = pd.read_csv('demand_lambda_dow.csv').set_index('dow')['factor']
precinct_probs = pd.read_csv('demand_lambda_precinct.csv').set_index('precinct')['pct_
of_total'] / 100

BASE_RATE = 3.482  # crashes per hour

def get_arrival_rate(hour: int, dow: int, area: str = 'all') -> float:
    """
    Get arrival rate λ(t) for given time.

    Args:
        hour: Hour of day (0-23)
        dow: Day of week (0=Monday, 6=Sunday)
        area: 'all', 'cbd', or 'non_cbd'

    Returns:
        Arrival rate in crashes per hour
    """
    rate = BASE_RATE * hourly_factors[hour] * dow_factors[dow]

    if area == 'cbd':
        rate *= 0.557  # CBD proportion
    elif area == 'non_cbd':
        rate *= 0.443  # Non-CBD proportion

    return rate

def generate_next_arrival(current_time: float, dt: float = 1.0) -> float:
    """
    Generate next arrival time using thinning algorithm.

    Args:
        current_time: Current simulation time (hours from start)
        dt: Time step for rate evaluation

    Returns:
        Time until next arrival (hours)
    """
    # Get maximum rate in next period
    hour = int(current_time % 24)
    dow = int((current_time // 24) % 7)
    lambda_max = get_arrival_rate(hour, dow) * 1.1  # Safety margin

    t = 0
    while True:
        # Generate candidate arrival
        t += np.random.exponential(1 / lambda_max)

        # Accept/reject
        new_hour = int((current_time + t) % 24)
        new_dow = int(((current_time + t) // 24) % 7)
        lambda_t = get_arrival_rate(new_hour, new_dow)

        if np.random.random() < lambda_t / lambda_max:
            return t

def assign_precinct() -> int:
```

```
    """Randomly assign crash to precinct based on demand distribution."""
    return np.random.choice(precinct_probs.index, p=precinct_probs.values)
```

Validation Checks

Before simulation, verify:
1. Generated hourly counts match expected patterns (within 10%)
2. Day-of-week distribution matches historical (within 5%)
3. Precinct assignments match spatial distribution

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7. Model Limitations

1. **Stationarity assumption**: Factors assumed constant across years
2. **Independence**: Events assumed independent (no clustering)
3. **Weather effects**: Not modeled (could affect rates ±20%)
4. **Special events**: Holidays, major events not explicitly modeled
5. **Trend**: Long-term trend not captured (data spans 13+ years)

Recommendations for Future Work

1. Include seasonal (monthly) factors
2. Add weather covariates
3. Model holiday effects
4. Consider Hawkes process for event clustering

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8. Output Files

File	Location	Description
`demand_lambda_hourly.csv`	`data/processed/`	24 hourly rates with CBD/non-CBD factors
`demand_lambda_dow.csv`	`data/processed/`	7 day-of-week factors
`demand_lambda_precinct.csv`	`data/processed/`	Precinct-level rates
`demand_model_summary.json`	`data/processed/`	Model summary statistics
`fig_demand_model_fit.png`	`results/figures/`	Diagnostic plots
`fig_hourly_rates.png`	`results/figures/`	CBD vs Non-CBD comparison
`fig_precinct_demand.png`	`results/figures/`	Precinct demand bar chart

References

1. Law, A.M. (2015). Simulation Modeling and Analysis, 5th ed. McGraw-Hill.
2. Ross, S.M. (2014). Introduction to Probability Models, 11th ed. Academic Press.
3. NYC Open Data - Motor Vehicle Collisions (NYPD)